Calculs de limites comparées

A] 1) $u_n = e^n - n$

2) $v_n = n \times e^{-n}$

3) $w_n = e^n - n^2$

4) $z_n = e^n - n^2 + n$

B] $u_n = e^{2n} - n$

$v_n = e^{4n} - 3\sqrt{n}$

$w_n = e^{n^2} - n^2$

$z_n = e^{n^2} - n^2 - n$

C] $u_n = \dfrac{e^{n^2}}{\sqrt{n}}$

$v_n = \frac{e^{n^2}}{n}$

$w_n = (n+1)\, e^{-n} + n^2$

$z_n = n e^{-3n^2+1}$

A] 1) $+\infty - \infty$: indéterminé.

$u_n = n\left(\dfrac{\boldsymbol{e^n}}{\boldsymbol{n}} - 1\right)$

$\lim \frac{e^n}{n} = +\infty$ donc $u_n \longrightarrow +\infty \times (+\infty - 1) = +\infty$ (opérations sur limites)